Hyeon Woo Lee

607-379-5619 / hyeonwoo610@gmail.com / pyodo.github.io / *US Permanent Resident*

Specialized in building robust AI systems for data-scarce, high-stakes deployment environments.

EXPERIENCE

Senior Staff Data Scientist | GE Healthcare April 2026 ~ Current

- Technical lead for a team of 5 data scientists developing autonomous X-ray systems, with hands-on development of the core feature, auto FOV estimation, and contributing to regulatory documentation and study design.
- Architecting digital twin simulation environments on NVIDIA Isaac for autonomous X-ray systems, directing data requirements and outputs to reduce reliance on real clinical data.
- Designing a teacher-student semi-supervised domain adaptation approach to train models on simulated data, partnering with the simulation team to improve data quality.
- Driving cross-functional collaboration across GE HealthCare and NVIDIA technical partners, co-developing the simulation pipeline and establishing shared technical standards.

AI Research Engineer | Philips July 2020 ~ April 2026

- Lead AI product development for Philips Ultrasound systems (*Lumify*, *Compact 5000 series*, and *Flash 5100*), enhancing real-time diagnostic capabilities for 2D and 3D ultrasound (*510(k) clearance - K252558*)
- Developed and deployed real-time (20 ms) on-device detection and segmentation models (serving with ONNX, TFLite) for FAST trauma exam guidance and an automated reporting system.
- Developed a large language model (LLM) pipeline for instruction prompting, retrieval-augmented generation (RAG) and fine-tuning to standardize unstructured ECG free-text, leveraging vLLM and TensorRT-LLM for faster, optimized inference.
- Designed and trained a transformer-based self-supervised representation model for ECG signal classification.
- Built and maintained data engineering and model validation toolkits to support high-quality data curation and rigorous model evaluation workflow aligned with clinical and regulatory standards.
- Contributed to FDA 510(k) and De Novo submissions by participating Q-Sub meetings and validation study designs. Authored key documentation, including device descriptions, clinical study reports, and AI software summaries.

Scientific Data Engineer | Allen Institute, Cell Science group July 2019 ~ July 2020

- Developer of the *Allen Cell & Structure Segmenter*, an open-source toolkit for microscopic cell images, focusing on computer vision and iterative learning workflows.
- Developed deep learning-based mitotic cell pair detection model and worked on a generative model for 3D image transfer across magnifications and resolutions.

TECHNICAL SKILLS

- **Programming Languages**: Python, C/C++, SQL, HTML/CSS.
- **Frameworks & Libraries**: PyTorch, TensorFlow, HF Transformers, vLLM, OpenCV, ITK, Matplotlib.
- **Tools & Platforms**: Docker, AWS, CUDA, MATLAB, Git, Linux.
- **Model Optimization**: Quantization, Pruning, Distillation, Model Conversion (ONNX & TFLite & TensorRT)

EDUCATION

Cornell University, Ithaca, NY May 2019
Master of Engineering in Biomedical Engineering
Research: Semi-supervised image segmentation (Presented at NeurIPS ML4H). Advised by Dr. Mert Sabuncu

University of Rochester, Rochester, NY December 2017
Bachelor of Science in Biomedical Engineering
Specialization: Signal processing, Machine Learning

PUBLICATIONS

H. Lee*, L.Chen*, R. Mukaddim, N. Balaraju, et al.
"Object Detection with Adaptive Annotation Styles in Ultrasound Videos via Teacher-Student Mutual Learning"
Ultrasound in Medicine & Biology, In Review, 2026

H. Lee, D. Dzikowicz, J.An
"Standardizing 12-Lead ECG Free-Text Reports to SNOMED-CT Using Mid-Sized Language Models"
IEEE EMBC, Accepted, 2026.

H. Lee, J.An
"Hybrid Convolution-Transformer Network Integrating Time and Frequency Domains from 12-Lead ECG for Chagas Disease Detection"
Computing in Cardiology (CinC) 2025.

H. Lee, W. Shi, R. Al Mukaddim, et al,
"Deep Learning-Based Automated Detection of the Middle Cerebral Artery in Transcranial Doppler Ultrasound Examinations"
Ultrasound in Medicine & Biology, 2025, Volume 51. Issue 9. pp 1555-1563

H. Lee, M. Zahiri, G. Goutam, et al
"Automated Anatomical Feature Detection for Completeness of Abdominal FAST Exam"
IEEE International Ultrasonics Symposium (IUS), 2023, pp. 1-4.

H. Lee, N. Schnittke, J. Fincke, et all
"Artificial Intelligence model to identify organ features for guiding FAST ultrasound exam"
American College of Emergency Physicians (ACEP), Annals of Emergency Medicine 80.4 (2022): S19.

K.A. Gerbin, T. Grancharova, R. M. Donovan-Maiye, … , **H. Lee**, et al
"Cell states beyond transcriptomics: integrating structural organization and gene expression in hiPSC-derived cardiomyocytes"
Cell Systems, 2021, Volume 12. Issue 6. pp 680-687.

J. Chen, L. Ding, M. P. Viana, **H. Lee**, et al
"The Allen Cell and Structure Segmenter: a new open source toolkit for segmenting 3D intracellular structures in fluorescence microscopy images"
bioRxiv, 2020.

H.W. Lee, M. R. Sabuncu, and A. V. Dalca.
"Few Labeled Atlases are Necessary for Deep-Learning-Based Segmentation"
NeurIPS ML4H: Machine Learning for Health, 2019 [Acc. rate: 26.1%].

PATENTS

M.H. Ghani, **H. Lee**, J. Fincke, B.I Raju — US Patent 12329577, Granted 6/2025
"Ultrasound Imaging"

P.K. Patel, C.E. Haverstock, **H. Lee**, et al — EP 4712868 A1, Published 3/2026
"Dynamic Determination of Imaging Sequence Completeness"

H. Lee, M.H. Ghani, J. Fincke, B.I Raju — US 20250359850 A1, Published 11/2025
"Supplemented Ultrasound"

H. Lee, G. Ghoshal, M. Zahiri, B.I Raju — US 20250077044 A1, Published 3/2025
"Systems and Methods of Personalized, Semi-Automatic Feature Analysis for Medical Imaging"

C.E. Haverstock, M. Xun, J. Fincke, P. Patel, … , **H. Lee** — US 20230320694 A1, Published 10/2023
"Graphical User Interface for Providing Ultrasound Imaging Guidance"